

COMSATS University Islamabad, Lahore Campus



Department of Computer Engineering
CPE314 – Data Communication & Computer Networks

Sakura Corp – Enterprise Network Design

Complex Engineering Assignment – Spring 2026

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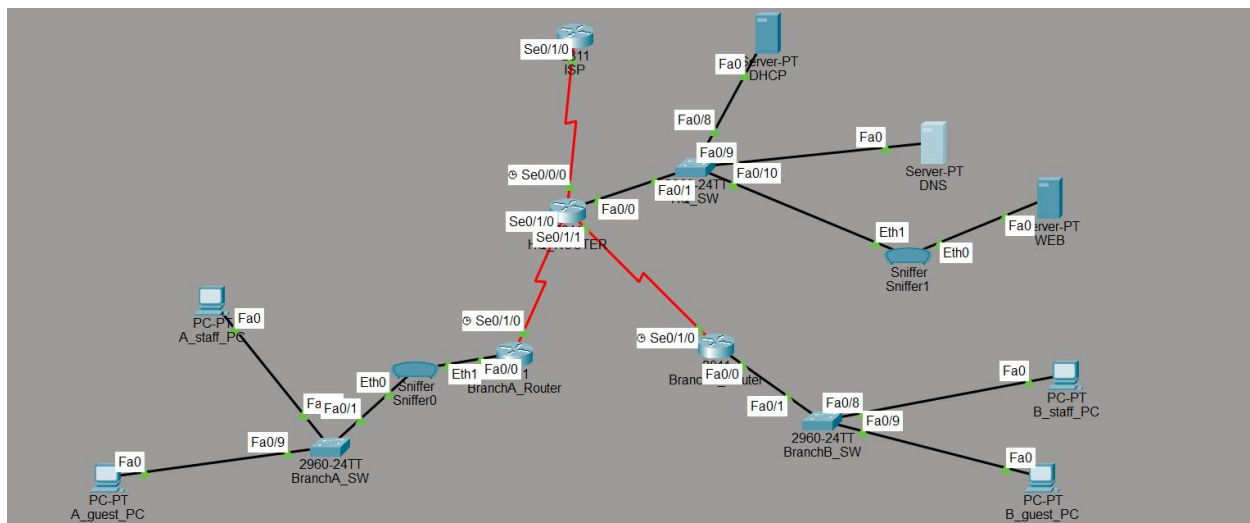
Instructor: Modassir Ishfaq | Date: 20-05-2026

1. Introduction

This report documents the complete network design and implementation for Sakura Corp, a growing organization with operations spread across three locations: a Headquarters (HQ) and two regional branches, Branch A and Branch B. As the Lead Network Architect, our goal was to build a secure, reliable, and centrally managed network that allows both branches to use services hosted at HQ while keeping their traffic isolated and controlled.

The design covers VLAN segmentation, centralized DHCP and DNS services, dynamic routing using OSPF, Router-on-a-Stick for inter-VLAN routing, Layer 2 security through port security and VTP, and Access Control Lists to enforce traffic policies between staff and guest users.

2. Network Architecture Overview



2.1 Headquarters (HQ)

The HQ acts as the central service hub for the entire organization. It hosts a dedicated Server Farm under VLAN 100 that provides three key services to all sites:

- DHCP Server (192.168.1.2): Assigns IP addresses to all PCs across HQ, Branch A, and Branch B using centralized pools and relay agents.
- DNS Server (192.168.1.3): Resolves the domain name `www.sakuracorp.local` to the Web Server IP address 192.168.1.4.
- HTTP Web Server (192.168.1.4): Hosts the Sakura Corp corporate portal with a customized `index.html` page containing student names and IDs.

DHCP DNS WEB SERVER

Static	Static	Static
192.168.1.2	192.168.1.3	192.168.1.4
255.255.255.0	255.255.255.0	255.255.255.0
192.168.1.1	192.168.1.1	192.168.1.1
192.168.1.3	192.168.1.3	192.168.1.3

2.2 Branch A

Branch A uses two VLANs on its local switch, both routed through the Branch A Router using Router-on-a-Stick:

- VLAN 10 (Staff): For employee traffic, using the subnet 192.168.10.0/24 with gateway 192.168.10.1.
- VLAN 20 (Guest): For visitor internet access, using the subnet 192.168.20.0/24 with gateway 192.168.20.1.

2.3 Branch B

Branch B follows the same VLAN structure as Branch A but uses completely different subnets to ensure OSPF can route between them:

- VLAN 10 (Staff): Subnet 192.168.30.0/24 with gateway 192.168.30.1.
- VLAN 20 (Guest): Subnet 192.168.40.0/24 with gateway 192.168.40.1.

3. VLSM Table

The following table shows the complete IP address plan for the Sakura Corp network. All LAN subnets use /24 masks providing 254 usable hosts each. WAN point-to-point links use /30 masks which give exactly 2 usable addresses per link, making efficient use of the address space.

Purpose	Location	Network	Subnet Mask	Gateway	Usable Range
Server Farm – VLAN 100	HQ	192.168.1.0/24	255.255.255.0	192.168.1.1	.2 – .254
Staff – VLAN 10	Branch A	192.168.10.0/24	255.255.255.0	192.168.10.1	.2 – .254
Guest – VLAN 20	Branch A	192.168.20.0/24	255.255.255.0	192.168.20.1	.2 – .254
Staff – VLAN 10	Branch B	192.168.30.0/24	255.255.255.0	192.168.30.1	.2 – .254

Purpose	Location	Network	Subnet Mask	Gateway	Usable Range
Guest – VLAN 20	Branch B	192.168.40.0/24	255.255.255.0	192.168.40.1	.2 – .254
WAN – HQ ↔ Branch A	WAN	192.168.200.0/30	255.255.255.252	–	.1 – .2
WAN – HQ ↔ Branch B	WAN	192.168.200.4/30	255.255.255.252	–	.5 – .6
WAN – HQ ↔ ISP	WAN	192.168.200.8/30	255.255.255.252	–	.9 – .10

4. Technical Implementation

4.1 VLAN and VTP Configuration

VLANs were created on each site's switch independently since VTP cannot cross a Layer 3 router. Each site operates in its own VTP domain:

- HQ Switch: VTP domain HQ, VLAN 100 (ServerFarm).

```
HQ_switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
100	ServerFarm	active	Fa0/8, Fa0/9, Fa0/10
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
HQ_switch#show vtp status
```

```
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : HQ
```

- Branch A Switch: VTP domain BranchA, VLAN 10 (Staff) and VLAN 20 (Guest).

```
BranchA_SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Staff	active	Fa0/8
20	Guest	active	Fa0/9
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
BranchA_SW#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : branchA
```

- Branch B Switch: VTP domain BranchB, VLAN 10 (Staff) and VLAN 20 (Guest).

```
BranchB_SW#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Staff	active	Fa0/8
20	Guest	active	Fa0/9
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
BranchB_SW#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
```

All switches run VTP Version 2 in Server mode. Access ports were assigned to their respective VLANs and all unused ports were shut down as a security measure. The uplink port on each switch connecting to the router was configured as an 802.1Q trunk.

4.2 Router-on-a-Stick (ROAS)

Since branches use multiple VLANs on a single router interface, Router-on-a-Stick was implemented at both Branch A and Branch B. Subinterfaces were created on the FastEthernet interface of each branch router, each with dot1Q encapsulation and its respective VLAN IP address as the default gateway for that VLAN.

```
interface FastEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.1.2
!
interface FastEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.1.2
```

```
interface FastEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.30.1 255.255.255.0
 ip helper-address 192.168.1.2
!
interface FastEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.40.1 255.255.255.0
 ip helper-address 192.168.1.2
```

At HQ, a single subinterface fa0/0.100 with encapsulation dot1Q 100 serves as the gateway for the Server Farm VLAN.

```
interface FastEthernet0/0.100
  encapsulation dot1Q 100
  ip address 192.168.1.1 255.255.255.0
```

4.3 Centralized DHCP with Relay Agents

The DHCP server at HQ (192.168.1.2) holds four address pools, one for each branch subnet. Since DHCP Discover messages are broadcasts that cannot cross routers by default, the ip helper-address command was applied to each branch subinterface pointing to 192.168.1.2. This causes the router to intercept the broadcast and forward it as a unicast to the DHCP server, filling in the giaddr field with its own subinterface IP so the server knows which pool to assign from.

Pool Name	Default Gateway	DNS Server	Start IP	Subnet Mask
BranchA_Staff	192.168.10.1	192.168.1.3	192.168.10.2	255.255.255.0
BranchA_Guest	192.168.20.1	192.168.1.3	192.168.20.2	255.255.255.0
BranchB_Staff	192.168.30.1	192.168.1.3	192.168.30.2	255.255.255.0
BranchB_Guest	192.168.40.1	192.168.1.3	192.168.40.2	255.255.255.0

Interface

FastEthernet0

Service

On

Off

Pool Name

serverPool

Default Gateway

0.0.0.0

DNS Server

0.0.0.0

Start IP Address :

192

168

1

0

Subnet Mask:

255

255

255

0

Maximum Number of Users :

512

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
BranchB_Guest	192.168.40.1	192.168.1.3	192.168.40.2	255.255.255.0	50	0.0.0.0	0.0.0.0
BranchB_Staff	192.168.30.1	192.168.1.3	192.168.30.2	255.255.255.0	50	0.0.0.0	0.0.0.0
BranchA_Guest	192.168.20.1	192.168.1.3	192.168.20.2	255.255.255.0	50	0.0.0.0	0.0.0.0
BranchA_Staff	192.168.10.1	192.168.1.3	192.168.10.2	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.168.1.0	255.255.255.0	512	0.0.0.0	0.0.0.0

4.4 OSPF Dynamic Routing

OSPF (Open Shortest Path First) was configured in Single Area 0 across all three routers (HQ, Branch A, Branch B) to ensure every subnet is reachable from everywhere. All directly connected networks were advertised using the network command with appropriate wildcard masks.

The HQ Router also has a default static route pointing to the ISP router (192.168.200.10), and this default route is redistributed into OSPF using the default-information originate command so that branch routers automatically learn the default gateway to the internet.

```
HQ_Router#show ip route ospf
O    192.168.10.0 [110/65] via 192.168.200.2, 02:38:43, Serial0/1/0
O    192.168.20.0 [110/65] via 192.168.200.2, 02:38:43, Serial0/1/0
O    192.168.30.0 [110/65] via 192.168.200.6, 02:38:43, Serial0/1/1
O    192.168.40.0 [110/65] via 192.168.200.6, 02:38:43, Serial0/1/1

BranchA_Router#show ip route ospf
O    192.168.1.0 [110/65] via 192.168.200.1, 02:36:25, Serial0/1/0
O    192.168.30.0 [110/129] via 192.168.200.1, 02:36:25, Serial0/1/0
O    192.168.40.0 [110/129] via 192.168.200.1, 02:36:25, Serial0/1/0
    192.168.200.0/24 is variably subnetted, 4 subnets, 2 masks
O        192.168.200.4 [110/128] via 192.168.200.1, 02:36:25, Serial0/1/0
O        192.168.200.8 [110/128] via 192.168.200.1, 02:36:25, Serial0/1/0
O*E2 0.0.0.0/0 [110/1] via 192.168.200.1, 02:36:25, Serial0/1/0

BranchB_Router#show ip route ospf
O    192.168.1.0 [110/65] via 192.168.200.5, 02:39:16, Serial0/1/0
O    192.168.10.0 [110/129] via 192.168.200.5, 02:39:06, Serial0/1/0
O    192.168.20.0 [110/129] via 192.168.200.5, 02:39:06, Serial0/1/0
    192.168.200.0/24 is variably subnetted, 4 subnets, 2 masks
O        192.168.200.0 [110/128] via 192.168.200.5, 02:39:16, Serial0/1/0
O        192.168.200.8 [110/128] via 192.168.200.5, 02:39:16, Serial0/1/0
O*E2 0.0.0.0/0 [110/1] via 192.168.200.5, 02:39:16, Serial0/1/0
```

Router	OSPF Networks Advertised
HQ_Router	192.168.1.0/24, 192.168.200.0/30, 192.168.200.4/30, 192.168.200.8/30
BranchA_Router	192.168.10.0/24, 192.168.20.0/24, 192.168.200.0/30
BranchB_Router	192.168.30.0/24, 192.168.40.0/24, 192.168.200.4/30

```
HQ_Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.200.2	0	FULL/ -	00:00:34	192.168.200.2	Serial0/1/0
192.168.200.6	0	FULL/ -	00:00:34	192.168.200.6	Serial0/1/1

```
BranchA_Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.200.9	0	FULL/ -	00:00:33	192.168.200.1	Serial0/1/0

```
BranchB_Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.200.9	0	FULL/ -	00:00:36	192.168.200.5	Serial0/1/0

4.5 Router Interface IP Addresses

Router	Interface	IP Address	Purpose
HQ_Router	Fa0/0.100	192.168.1.1	VLAN 100 gateway
HQ_Router	Se0/1/0	192.168.200.1	WAN to Branch A
HQ_Router	Se0/1/1	192.168.200.5	WAN to Branch B
HQ_Router	Se0/0/0	192.168.200.9	WAN to ISP
BranchA_Router	Fa0/0.10	192.168.10.1	VLAN 10 gateway
BranchA_Router	Fa0/0.20	192.168.20.1	VLAN 20 gateway
BranchA_Router	Se0/1/0	192.168.200.2	WAN to HQ
BranchB_Router	Fa0/0.10	192.168.30.1	VLAN 10 gateway
BranchB_Router	Fa0/0.20	192.168.40.1	VLAN 20 gateway
BranchB_Router	Se0/1/0	192.168.200.6	WAN to HQ
ISP Router	Se0/1/0	192.168.200.10	WAN to HQ

4.6 Layer 2 Security – Port Security

Port security was enabled on all access ports across all three switches to prevent unauthorized devices from connecting to the network. The configuration used:

- Maximum MAC addresses: 2 (allows one device with some flexibility)
- Violation mode: Shutdown (port is disabled if a violation is detected)

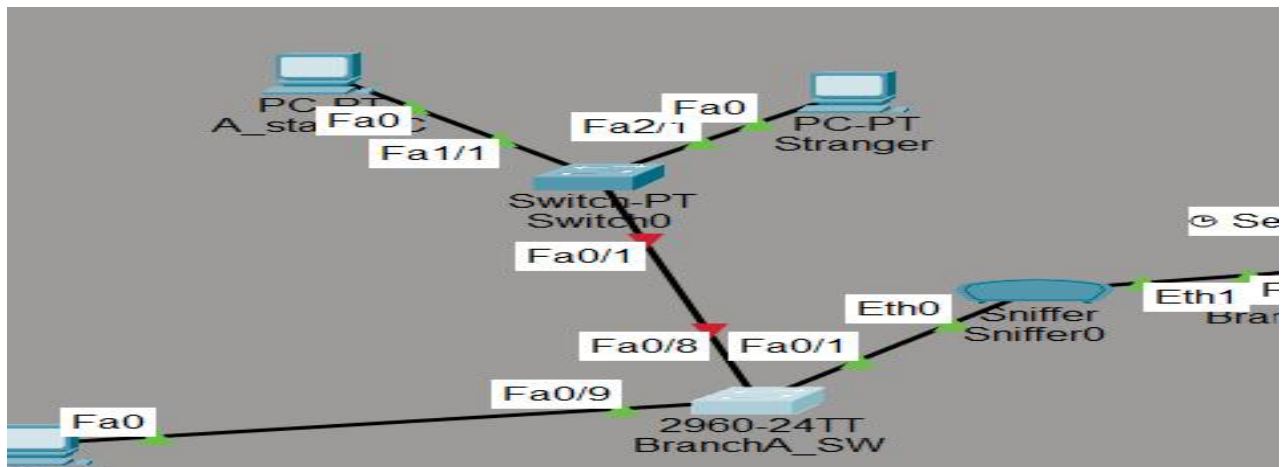
```
BranchA_SW#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/8      2              1              0          Shutdown
Fa0/9      2              1              0          Shutdown
-----
```

```
BranchB_SW#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/8      2              1              0          Shutdown
Fa0/9      2              1              0          Shutdown
-----
```

```
HQ_switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/8      2              1              0          Shutdown
Fa0/9      2              0              0          Shutdown
Fa0/10     2              0              0          Shutdown
-----
```

Trunk ports were excluded from port security as they carry traffic for multiple VLANs. All unused ports were also administratively shut down.

Verifying Violation:



```
BranchA_SW#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/8      2              0              1          Shutdown
Fa0/9      2              1              0          Shutdown
-----
```

```
%PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address
0001.C985.39B8 on port FastEthernet0/8.
```

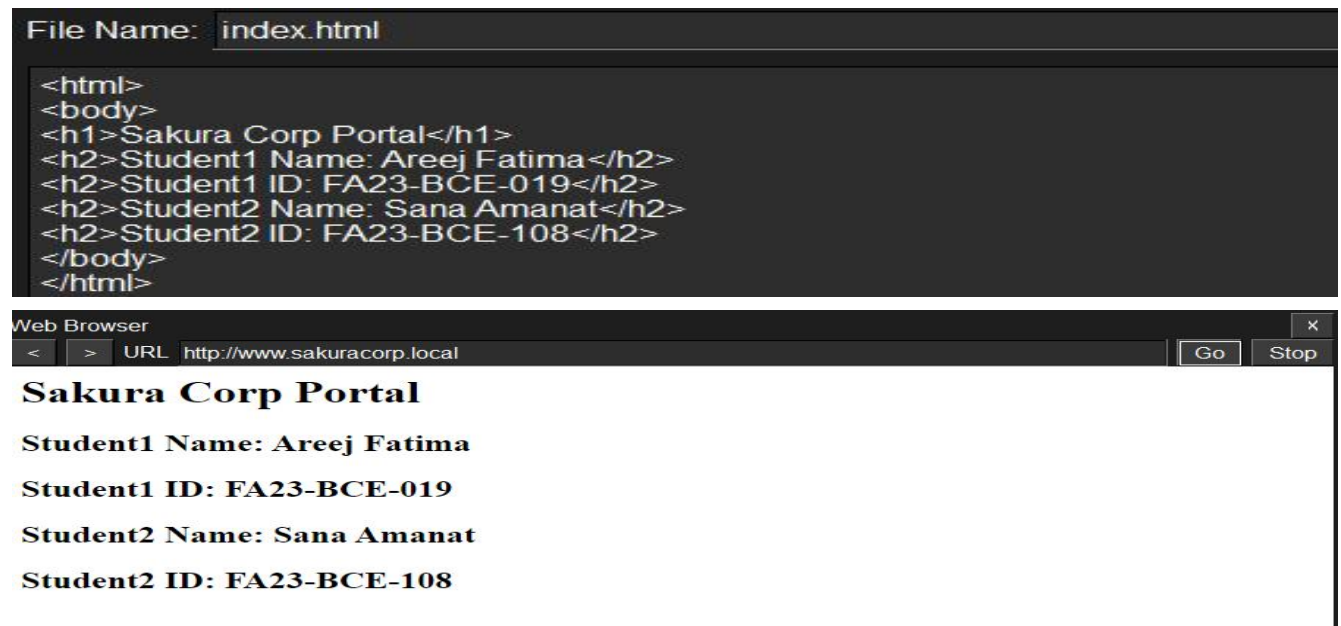
```
Switch#show mac-address
Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1     0001.c985.39b8    DYNAMIC   Fa2/1
```

4.7 DNS and Web Server

The DNS Server at 192.168.1.3 was configured with a single A Record mapping `www.sakuracorp.local` to 192.168.1.4 (the Web Server IP). This allows any PC in the network to resolve the corporate portal by domain name rather than IP address.

DNS			
DNS Service		☒ On	☐ Off
Resource Records			
Name	www.sakuracorp.local		Type A Record
Address	192.168.1.4		
Add		Save	Remove
No.	Name	Type	Detail
0	www.sakuracorp.local	A Record	192.168.1.4

The Web Server at 192.168.1.4 hosts the corporate portal. The `index.html` file was customized to display the student names and registration numbers as required by the assignment.



4.8 Access Control Lists

Extended Named ACLs were applied inbound on the Guest subinterfaces of both Branch A and Branch B routers. The policy enforces the following rules:

- Guest users (VLAN 20) are permitted to access the Web Server on port 80 (HTTP) only.
- Guest users are blocked from sending ICMP (ping) to any device in the Server Farm (192.168.1.0/24).
- Staff users (VLAN 10) have full unrestricted access to all resources including the Server Farm.

```
BranchA_Router#show access-lists
Extended IP access list GUEST_POLICY
 10 permit tcp 192.168.20.0 0.0.0.255 192.168.1.0 0.0.0.255 eq www
 20 deny icmp 192.168.20.0 0.0.0.255 192.168.1.0 0.0.0.255
 30 permit ip any any (3 match(es))

BranchB_Router#show access-lists
Extended IP access list GUEST_POLICY
 10 permit tcp 192.168.40.0 0.0.0.255 192.168.1.0 0.0.0.255 eq www
 20 deny icmp 192.168.40.0 0.0.0.255 192.168.1.0 0.0.0.255
 30 permit ip any any (2 match(es))
```

The ACL named GUEST_POLICY was applied to fa0/0.20 in the inbound direction on both branch routers. Placing the ACL on the branch routers close to the source is consistent with best practices for extended ACLs.

Branch A guest PC:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Branch B staff PC:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=8ms TTL=126
Reply from 192.168.1.2: bytes=32 time=13ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 5ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.4: bytes=32 time=16ms TTL=126
Reply from 192.168.1.4: bytes=32 time=2ms TTL=126
Reply from 192.168.1.4: bytes=32 time=26ms TTL=126

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 26ms, Average = 14ms
```

Branch B guest PC:

Web Browser

< >

URL http://www.sakuracorp.local

Go Stop

Sakura Corp Portal

Student1 Name: Areej Fatima

Student1 ID: FA23-BCE-019

Student2 Name: Sana Amanat

Student2 ID: FA23-BCE-108

5. Analysis and Diagnostic Report

5.1 DHCP Relay Analysis – giaddr Field

To verify that the DHCP relay mechanism was working correctly, a sniffer was placed on the link between the Branch A Switch and Branch A Router. When A_staff_PC sent a DHCP Discover, the packet was captured and analyzed.

The key observation is in the IP header of the forwarded packet where the source IP shows 192.168.10.1, which is the Branch A Router subinterface IP. This is the giaddr (Gateway IP Address) field that the router inserts when it relays the broadcast to the DHCP server.

The role of giaddr in subnet identification is critical. When the HQ DHCP server receives the relayed request, it reads the giaddr field to determine which subnet the client belongs to. In this case, 192.168.10.1 tells the server the request came from the 192.168.10.0/24 subnet, so it assigns an address from the BranchA_Staff pool. Without giaddr, the DHCP server would have no way of knowing which pool to use and the relay mechanism would fail.

The captured DHCP ACK packet also confirmed the assignment was correct, showing YOUR CLIENT ADDRESS: 192.168.10.8 and SERVER ADDRESS: 192.168.1.2, proving end-to-end relay functionality.

PC sends DHCP Discover :

When A_staff_PC is set to DHCP, it has no IP address yet. It sends a DHCP Discover as a broadcast with source IP 0.0.0.0 and destination 255.255.255.255. This packet is captured on the switch side (Port0) before it reaches the router. At this point the giaddr field is empty because the relay has not processed it yet.

The image shows two windows from a network simulation environment.

A_staff_PC Configuration:

- Physical:** FastEthernet0
- IP Configuration:**
 - Interface: FastEthernet0
 - IP Configuration: DHCP (selected), Static (unselected)
 - IPv4 Address: 192.168.10.8
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
 - DNS Server: 192.168.1.3
- IPv6 Configuration:**
 - Automatic (selected), Static (unselected)
 - IPv6 Address: (empty)
 - Link Local Address: FE80::201:63FF:FE07:5764
 - Default Gateway: (empty)
 - DNS Server: (empty)
- 802.1X:**
 - Use 802.1X Security: (unchecked)
 - Authentication: MD5
 - Username: (empty)
 - Password: (empty)

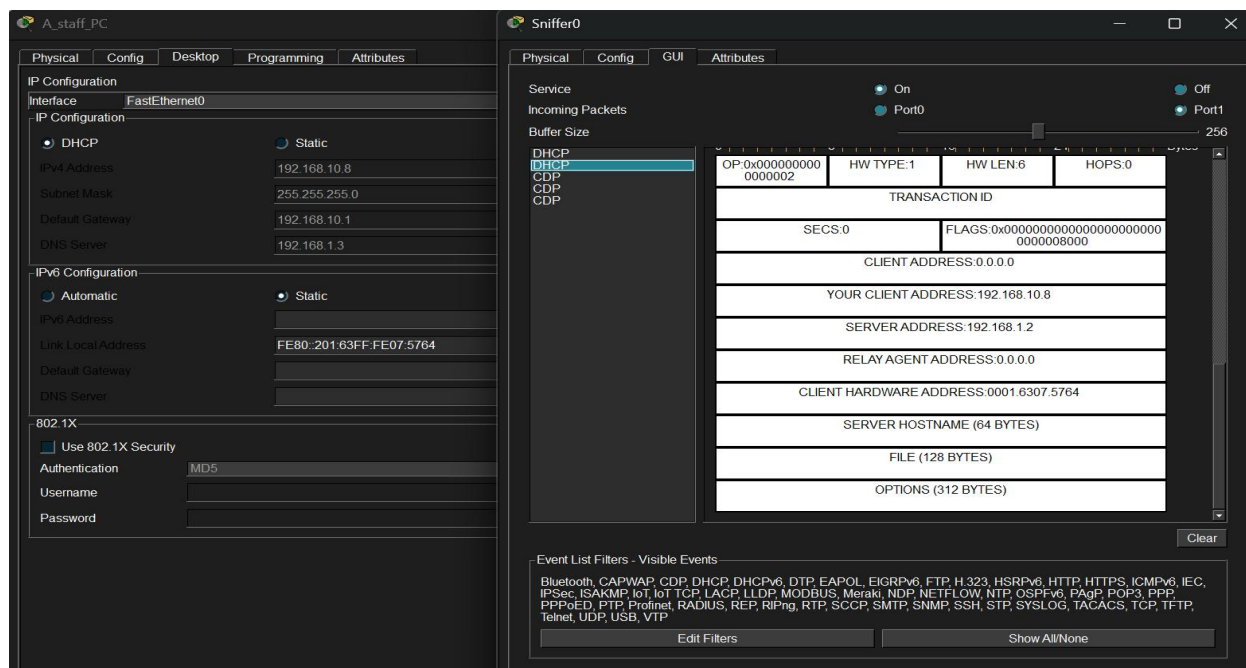
Sniffer0 Configuration and Capture:

- Physical:** Port0
- Service:** On
- Incoming Packets:** Port0
- Buffer Size:** 256
- Event List:** DHCP (selected)
- Packet Details:**
 - Ethernet II:** Preamble: 101010...10, SF: 0, DEST ADDR: FFFF.FFFF.FFFF, SRC ADDR: 0001.6307.5764, TPID: 0x8100, TCI: 0x000a, Type: 0x0001.
 - IP:** VER: 4, IHL: 5, DSCP: 0x00, TL: 77, ID: 0x0038, FLAG: S: 0x0, FRAG OFFSET: 0x000, TTL: 128, PRO: 0x11, CHKSUM: (empty), SRC IP: 0.0.0.0, DST IP: 255.255.255.255, DATA: (VARIABLE LENGTH).

Router processes and relays:

The same DHCP Discover packet is now captured on the router side (Port1) after the Branch A Router processed it. The source IP has changed from 0.0.0.0 to 192.168.10.1, which is the Branch A Router's fa0/0.10 subinterface IP. This IP is inserted into the giaddr field by the ip helper-address mechanism. The router then forwards this packet as a unicast to the HQ DHCP server at 192.168.1.2. The giaddr value 192.168.10.1 tells the DHCP server that the request originated from the 192.168.10.0/24 subnet so it must assign an address from the BranchA_Staff pool.

The image shows two windows from a network simulation. The left window, titled 'A_staff_PC', displays the configuration for the 'FastEthernet0' interface. Under 'IP Configuration', the 'DHCP' option is selected, and the 'Static' option is also checked. The IPv4 Address is set to 192.168.10.8, Subnet Mask to 255.255.255.0, Default Gateway to 192.168.10.1, and DNS Server to 192.168.1.3. The IPv6 Configuration is set to 'Automatic'. The '802.1X' section shows 'Use 802.1X Security' is checked, with 'Authentication' set to 'MD5'. The right window, titled 'Sniffer0', shows a packet capture of a DHCP Discover packet. The packet is captured on 'Ethernet802.1q' and is a DHCP Discover packet. The packet details show: PREAMBLE: 101010.10, SRC ADDR: 0007.EC34.7D01, TPID: 0x8100, TC: 0x000a, Type: 0x1, DEST ADDR: FFFF.FFFF.FFFF, DATA (VARIABLE LENGTH), FCS: 0x00000000. The packet is also shown in hex and ASCII format. The 'Event List Filters - Visible Events' section lists various network protocols, including Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DTP, EAPOL, EIGRPv6, FTP, H.323, HSRPv6, HTTP, HTTPS, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIPv2, RIPv3, RIPv4, RIPv5, RIPv6, RIPv7, RIPv8, RIPv9, RIPv10, RIPv11, RIPv12, RIPv13, RIPv14, RIPv15, RIPv16, RIPv17, RIPv18, RIPv19, RIPv20, RIPv21, RIPv22, RIPv23, RIPv24, RIPv25, RIPv26, RIPv27, RIPv28, RIPv29, RIPv30, RIPv31, RIPv32, RIPv33, RIPv34, RIPv35, RIPv36, RIPv37, RIPv38, RIPv39, RIPv40, RIPv41, RIPv42, RIPv43, RIPv44, RIPv45, RIPv46, RIPv47, RIPv48, RIPv49, RIPv50, RIPv51, RIPv52, RIPv53, RIPv54, RIPv55, RIPv56, RIPv57, RIPv58, RIPv59, RIPv60, RIPv61, RIPv62, RIPv63, RIPv64, RIPv65, RIPv66, RIPv67, RIPv68, RIPv69, RIPv70, RIPv71, RIPv72, RIPv73, RIPv74, RIPv75, RIPv76, RIPv77, RIPv78, RIPv79, RIPv80, RIPv81, RIPv82, RIPv83, RIPv84, RIPv85, 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5.2 TCP 3-Way Handshake Analysis

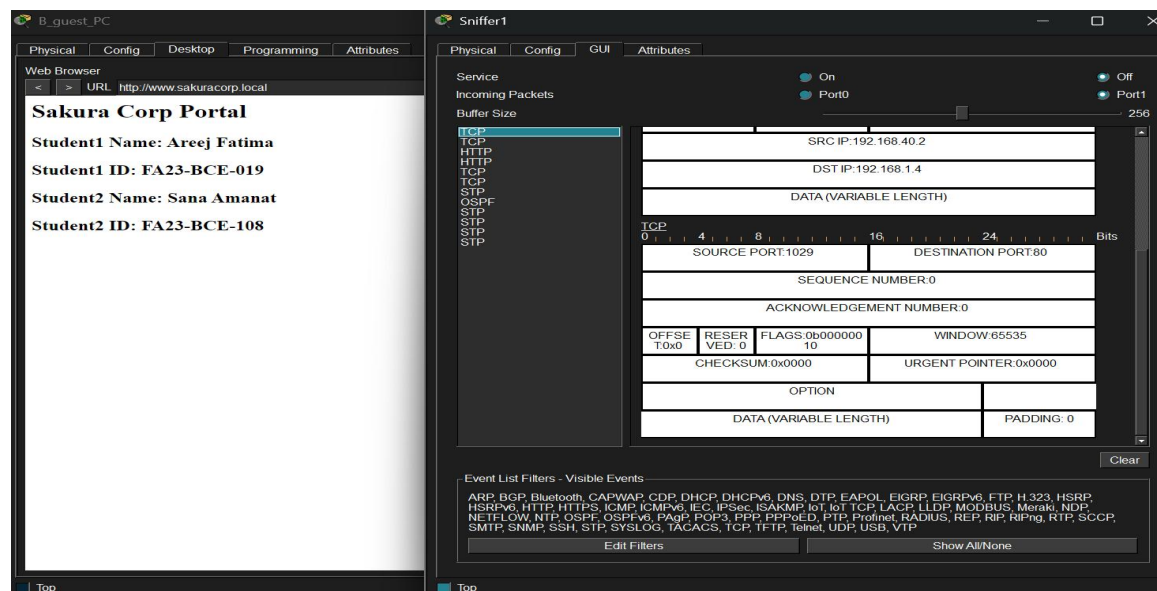
To capture the TCP handshake, a sniffer was placed between HQ Switch and the Web Server. B_guest_PC accessed www.sakuracorp.local through the web browser, and the following three packets were captured:

Step	Packet	Source IP	Destination IP	FLAGS	Description
1	SYN	192.168.40.2	192.168.1.4	0b00000010	B_guest_PC initiates a TCP connection to Web Server on port 80. Only the SYN flag is set. Sequence number is 0.
2	SYN-ACK	192.168.1.4	192.168.40.2	0b00010010	Web Server accepts the connection. Both SYN and ACK flags are set. Server acknowledges the client sequence number.
3	ACK	192.168.40.2	192.168.1.4	0b00010000	B_guest_PC confirms the connection. Only ACK flag is set. TCP connection is now fully established and HTTP data transfer begins.

The successful TCP handshake from Branch B to the HQ Web Server confirms that OSPF routing, DHCP relay, DNS resolution, and HTTP access are all functioning correctly together across the enterprise network.

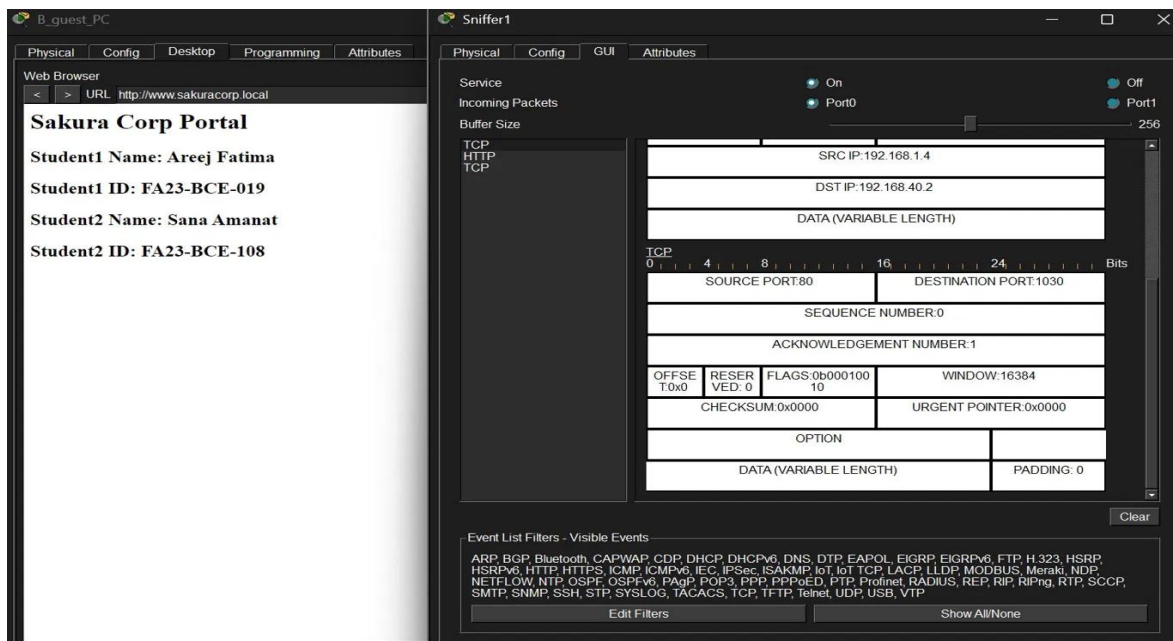
SYN:

B_guest_PC at 192.168.40.2 sends a SYN packet to the Web Server at 192.168.1.4 on port 80. Only the SYN flag is set (0b00000010). Sequence number is 0 and acknowledgement number is 0 because no data has been exchanged yet. This is the first step of the TCP 3-way handshake where the client requests a connection



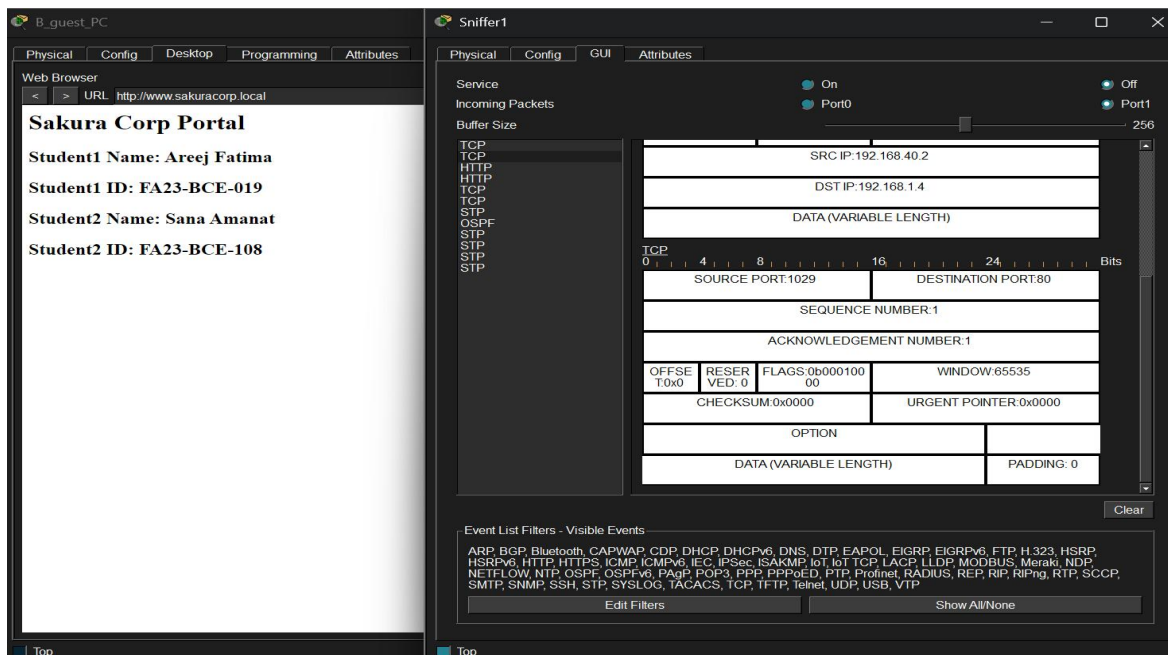
SYN-ACK:

The Web Server at 192.168.1.4 responds with a SYN-ACK packet. Both SYN and ACK flags are set (0b00010010). The acknowledgement number is 1 confirming it received the client SYN. The server sets its own sequence number to 0 to synchronize with the client. This is the second step confirming the server is ready to accept the connection



ACK:

B_guest_PC sends the final ACK packet completing the handshake. Only the ACK flag is set (0b00010000). Sequence number is 1 and acknowledgement number is 1 confirming both sides are synchronized. The TCP connection is now fully established and HTTP data transfer begins. The Sakura Corp portal loaded successfully in the browser confirming end-to-end connectivity from Branch B to HQ Web Server



6. Conclusion

The Sakura Corp enterprise network was successfully designed and implemented in Cisco Packet Tracer. The network provides centralized IP management through DHCP relay, dynamic routing via OSPF, secure Layer 2 configurations with port security and VTP, and enforced access control policies separating guest and staff traffic.

All branch PCs received correct IP addresses from the HQ DHCP server, the DNS server correctly resolved the corporate domain, and the Web Server was accessible from both branches by domain name. Guest users were successfully restricted to HTTP-only access while being blocked from pinging the Server Farm, and staff users retained full access as required.

The Wireshark captures using Packet Tracer's sniffer tool provided evidence of the DHCP relay mechanism through the giaddr field and the TCP 3-way handshake confirming end-to-end connectivity across the routed enterprise network.